INCH-POUND

MS51504C
w/AMENDMENT 1
20 January 2017
SUPERSEDING
MS51504C
7 June 2016

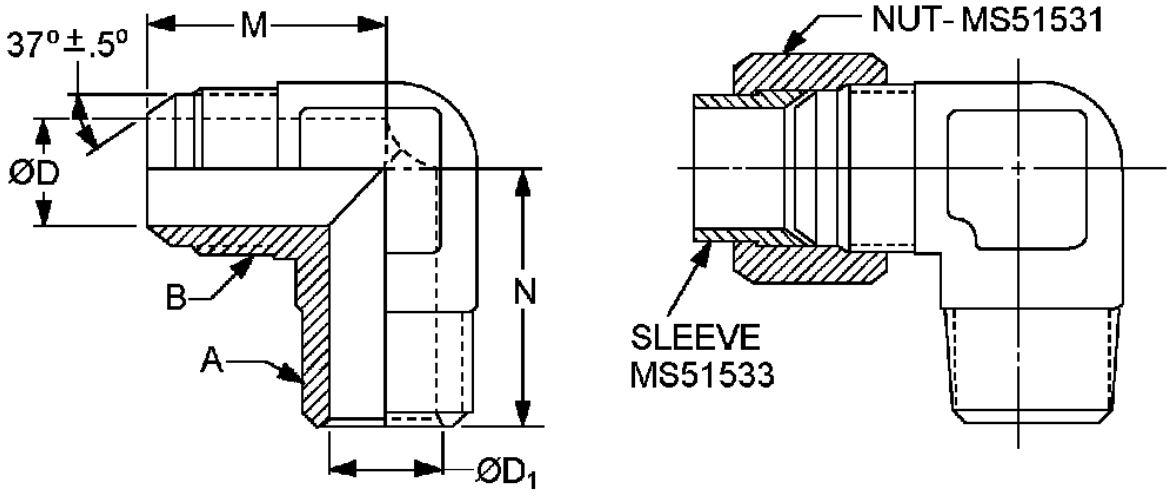
DETAIL SPECIFICATION SHEET

ELBOW, PIPE TO TUBE, 90 DEGREE
MALE PIPE END, 37 DEGREE FLARED

Reactivated after 20 January 2017 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. 90° male elbow.

AMSC N/A

FSC 4730

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Suffix designator		Tube OD nom. inches (mm)	A Pipe thread NPTF	B Straight thread
Assembly (see note 4)	Body			
A2	B2	.1250 (3.175)	.125 -27	.3125-24 UNF-2A
A3	B3	.1875 (4.763)	.125 -27	.3750-24 UNF-2A
A4	B4	.2500 (6.350)	.125 -27	.4375-20 UNF-2A
A4-4	B4-4	.2500 (6.350)	.250 - 18	.4375-20 UNF-2A
A5	B5	.3125 (7.938)	.125 -27	.5000-20 UNF-2A
A5-4	B5-4	.3125 (7.938)	.250 - 18	.5000-20 UNF-2A
A6-2	B6-2	.3750 (9.525)	.125 -27	.5625-18 UNF-2A
A6	B6	.3750 (9.525)	.250 - 18	.5625-18 UNF-2A
A6-6	B6-6	.3750 (9.525)	.375 - 18	.5625-18 UNF-2A
A6-8	B6-8	.3750 (9.525)	.500 - 14	.5625-18 UNF-2A
A8-4	B8-4	.5000 (12.700)	.250 - 18	.7500-16 UNF-2A
A8	B8	.5000 (12.700)	.375 - 18	.7500-16 UNF-2A
A8-8	B8-8	.5000 (12.700)	.500 - 14	.7500-16 UNF-2A
A8-12	B8-12	.5000 (12.700)	.750 - 14	.7500-16 UNF-2A
A10-6	B10-6	.6250 (15.875)	.375 - 18	.8750-14 UNF-2A
A10	B10	.6250 (15.875)	.500 - 14	.8750-14 UNF-2A
A12-8	B12-8	.7500 (19.050)	.500 - 14	1.0625-12 UN-2A
A12	B12	.7500 (19.050)	.750 - 14	1.0625-12 UN-2A
A14	B14	.8750 (22.225)	.750 - 14	1.1875-12 UN-2A
A16-12	B16-12	1.0000 (25.400)	.750 - 14	1.3125-12 UN-2A
A16	B16	1.0000 (25.400)	1.000 - 11 1/2	1.3125-12 UN-2A
A20	B20	1.2500 (31.750)	1.250 - 11 1/2	1.6250-12 UN-2A
A24	B24	1.5000 (38.100)	1.500 - 11 1/2	1.8750-12 UN-2A
A32	B32	2.0000 (50.800)	2.000 - 11 1/2	2.500-12 UN-2A

See notes at end of figure.

FIGURE 1. 90° male elbow - Continued.

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Suffix designator		D diameter		D_1 dia. drill inches (mm)	M inches (mm) ±.030 (0.76)	N inches (mm) ±.030 (0.76)
Assembly (see note 4)	Body	Basic inches (mm)	Tolerance inches (mm)			
A2	B2	.062 (1.57)	±.003 (0.08)	.188 (4.78)	.770 (19.56)	.720 (18.29)
A3	B3	.125 (3.18)		.188 (4.78)	.830 (21.08)	.720 (18.29)
A4	B4	.172 (4.37)		.188 (4.78)	.890 (22.61)	.780 (19.81)
A4-4	B4-4	.172 (4.37)		.281 (7.14)	1.030 (26.16)	1.090 (27.69)
A5	B5	.234 (5.94)		.188 (4.78)	.950 (24.13)	.780 (19.81)
A5-4	B5-4	.234 (5.94)		.281 (7.14)	1.050 (26.67)	1.090 (27.69)
A6-2	B6-2	.297 (7.54)	±.004 (0.10)	.188 (4.78)	1.060 (26.92)	.900 (22.86)
A6	B6	.297 (7.54)		.281 (7.14)	1.060 (26.92)	1.090 (27.69)
A6-6	B6-6	.297 (7.54)		.406 (10.31)	1.140 (28.96)	1.220 (30.99)
A6-8	B6-8	.297 (7.54)		.533 (13.54)	1.220 (30.99)	1.470 (37.34)
A8-4	B8-4	.391 (9.93)		.281 (7.14)	1.250 (31.75)	1.220 (30.99)
A8	B8	.391 (9.93)		.406 (10.31)	1.250 (31.75)	1.220 (30.99)
A8-8	B8-8	.391 (9.93)		.531 (13.49)	1.330 (33.78)	1.470 (37.34)
A8-12	B8-12	.391 (9.93)		.719 (18.26)	1.420 (36.07)	1.590 (40.39)
A10-6	B10-6	.484 (12.29)		.406 (10.31)	1.450 (36.83)	1.280 (32.51)
A10	B10	.484 (12.29)		.531 (13.49)	1.450 (36.83)	1.470 (37.34)
A12-8	B12-8	.609 (15.47)	±.005 (0.13)	.531 (13.49)	1.660 (42.16)	1.590 (40.39)
A12	B12	.609 (15.47)		.719 (18.26)	1.660 (42.16)	1.590 (40.39)
A14	B14	.718 (18.24)		.719 (18.26)	1.730 (43.94)	1.690 (42.93)
A16-12	B16-12	.844 (21.44)	+.007 -.005 (+0.18 -0.13)	.719 (18.26)	1.810 (45.97)	1.780 (45.21)
A16	B16	.844 (21.44)		.938 (23.83)	1.810 (45.97)	1.970 (50.04)
A20	B20	1.078 (27.38)	+.006 -.005 (+0.15 -0.13)	1.250 (31.75)	2.060 (52.32)	2.380 (60.45)
A24	B24	1.312 (33.32)		1.500 (38.10)	2.330 (59.18)	2.640 (67.06)
A32	B32	1.781 (45.24)	+.010 -.005 (+0.25 -0.13)	1.938 (49.23)	3.060 (77.72)	3.000 (76.20)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked, stored, or issued.
5. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
7. It is recommended that threads be dressed prior to assembly.

FIGURE 1. 90° male elbow - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in table I.

Components of the assemblies shall be of the same material and finish.

Materials shall be in accordance with MIL-DTL-18866.

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Finish. Finishes shall be in accordance with table I. All platings shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 3/ 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 5/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 5/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 4/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
3/ Hexavalent chromium free. Finish shall be ROHS compliant.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to FED-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

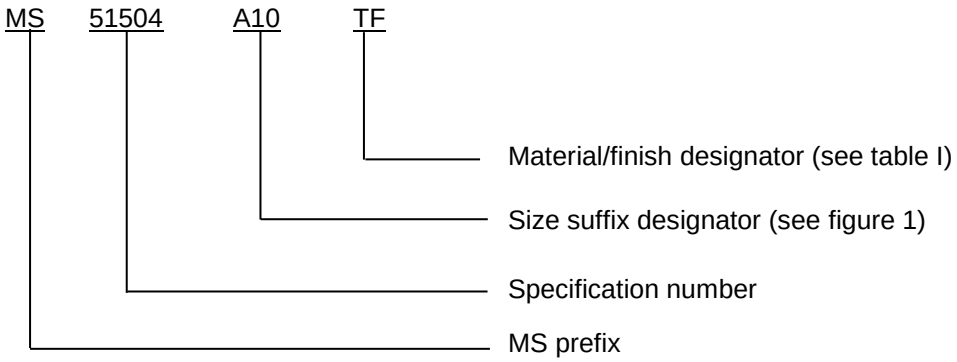
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to the withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

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PIN: The PIN consists of the letters “MS”, the specification sheet number, a letter and number for elbow size, and a letter for material finish designator.



PIN example: MS51504A10TF indicates an elbow 90°, pipe to tube, .6250 inch (15.875 mm), titanium fluoride phosphate.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment Notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-595/36076	FED-STD-595/36176	MS51533	SAE-AMS2451/5
FED-STD-595/36081	FED-STD-595/36231	ASTM B117	SAE-AMS2451/9
FED-STD-595/36099	FED-STD-595/36251	ASTM B633	SAE-AMS2486
FED-STD-595/36118	FED-STD-595/36270	ASTM B695	SAE-AMS2488
FED-STD-595/36134	FED-STD-595/36280	ASTM F1136/F1136M	SAE-AMS2700
FED-STD-595/36152	FED-STD-595/36293	SAE-AMS-C-81562	SAE-J514
FED-STD-595/36170	MIL-DTL-16232	SAE-AMS-QQ-P-416	
FED-STD-595/36173	MS51531	SAE-AMS2417	

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2017-009)

Review activities:
Army - AR, AV, MI
Navy - AS, CG, SA
Air Force - 71

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